

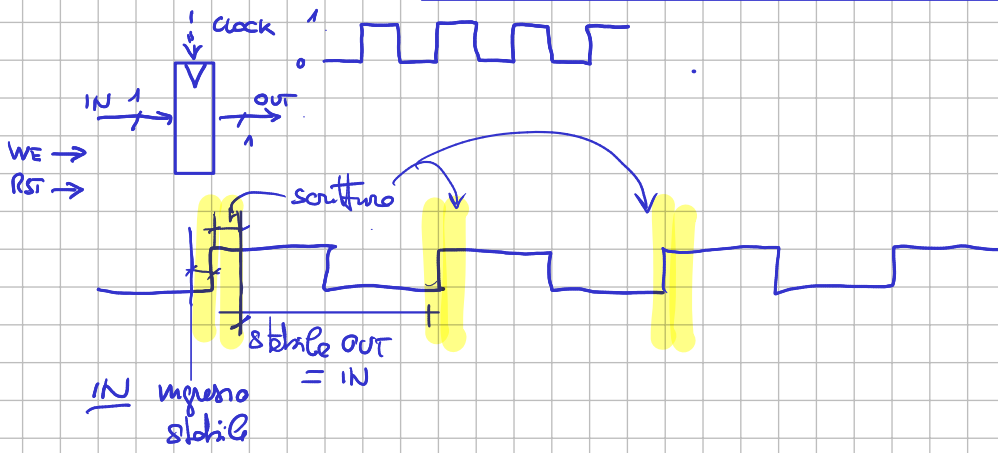
REGISTRO

RETI

LOGICHE

SEQ

: fun con stato



WE write enable

Scrittura solo quando WE = 1

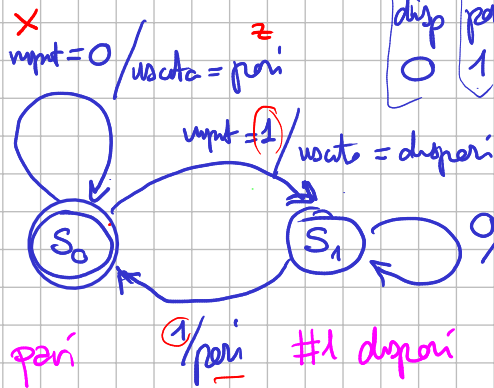
FUN CON STATO

↔

AUTOMI

porta di una seq di bit

↓
#1



tempo				
1	1	0	1	
disp	par	disp	disp	pari
0	1	0	0	1

FIND
di Moore

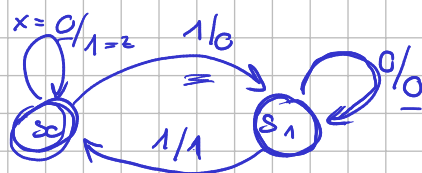
automa di
Mealy

- 1) stato $\begin{cases} S_0 \\ S_1 \end{cases}$
- 2) funzione che calcola le uscite $\begin{cases} \text{pari} \\ \text{dispari} \end{cases}$
- 3) funzione che calcola il prossimo stato interno

Procedimento Standard

- 1) automa
- 2) funz. uscite, funz. prossimo stato
- 3) stato
- 4) ciclo di clock

1)



$$s_0 = 0$$

$$s_1 = 1$$

3) Stato 2 valori
 $[q_2] = 1$ 1 bit

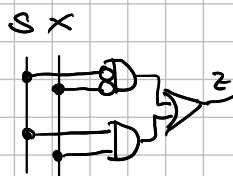


2) Funzione che calcola l'uscita a partire da S e X

S	X	Z
0	0	1
0	1	0
1	0	0
1	1	1

$$Z = \bar{S}\bar{X} + SX$$

S \ X	0	1
0	1	0
1	0	1

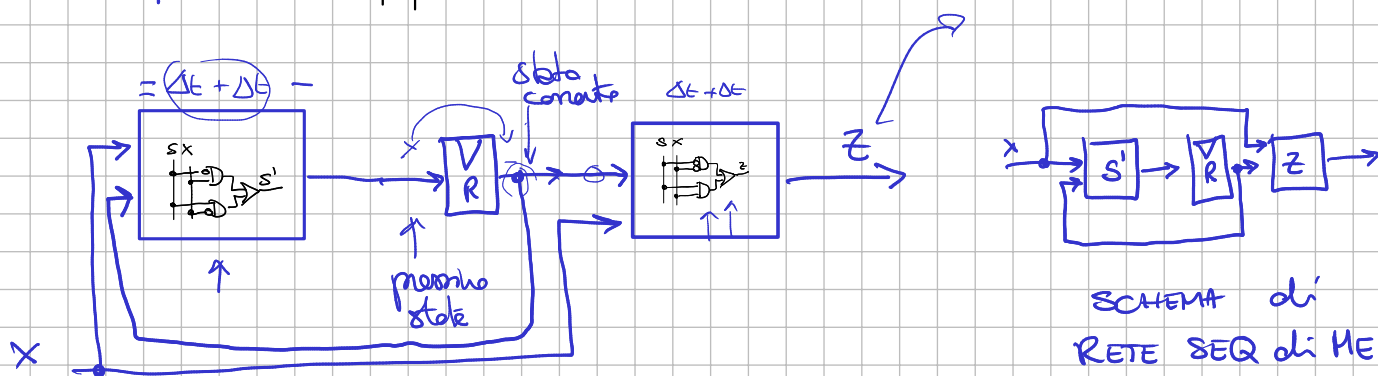
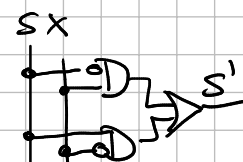


funzione che calcola il prossimo stato interno a partire da S e X

S	X	S'
0	0	0
0	1	1
1	0	1
1	1	0

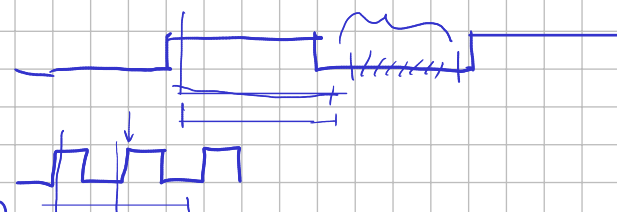
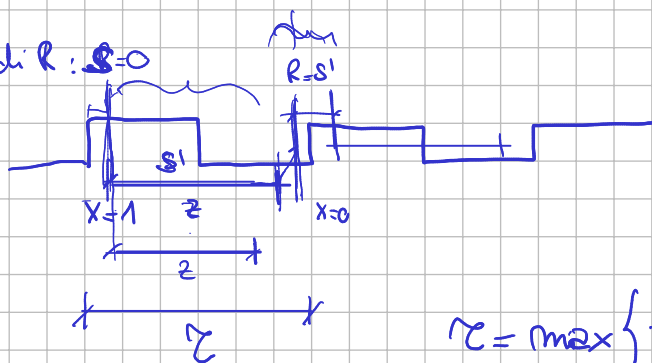
$$S' = \bar{S}X + S\bar{X}$$

S \ X	0	1
0	0	1
1	1	0

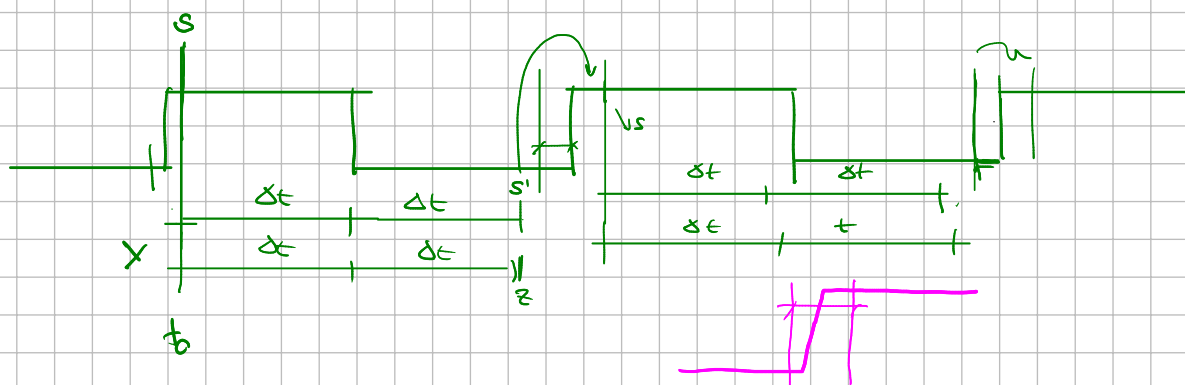


SCHEMA di RETE SEQ di MEALY

uscita di R: S=0



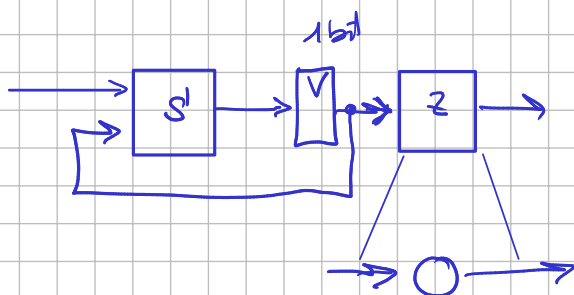
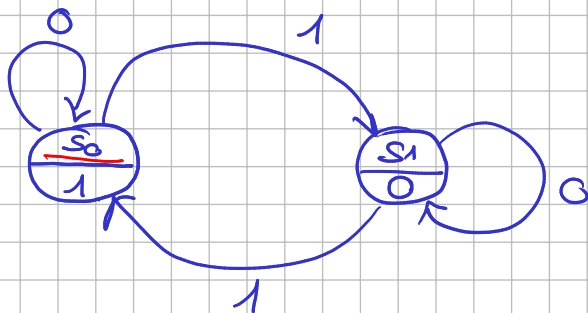
$$T_c = \max\{t_{s'}, t_z\} + t_{write}$$



Moore (automi dr)

$$z : \underline{S} \rightarrow z$$

portate di
seq di bit



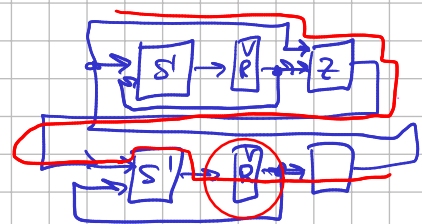
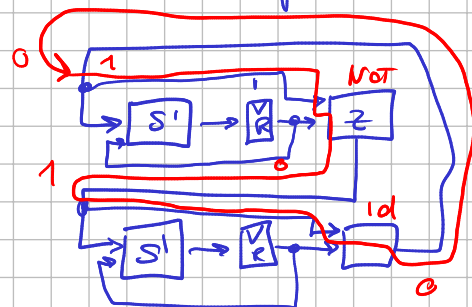
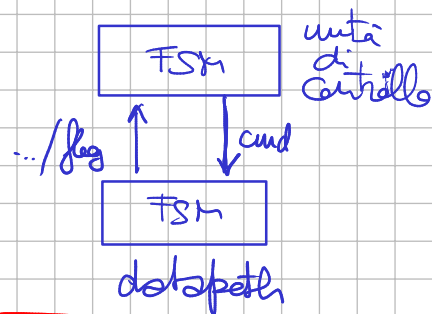
$$S_0 = 1$$

$$S_1 = 0$$

$$z = S$$

	cod std	funke std	cod
S_0	00		100
S_1	01	01	010
S_2	10	11	001

Processore



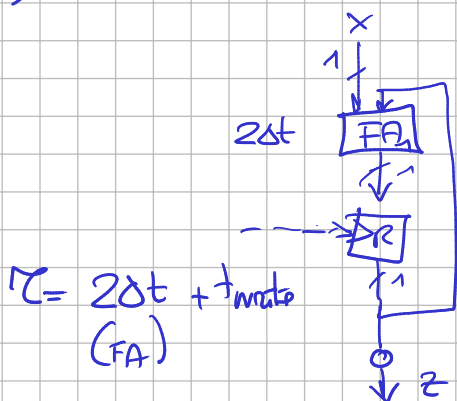
Moore

1) autome

2)

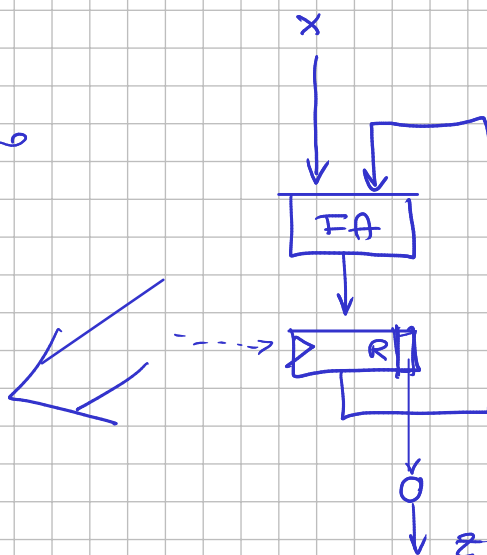
fun	uscite
fun	prossimo stato interno

3) registro stato



$$T = 2\delta t + t_{write}$$

(FA)



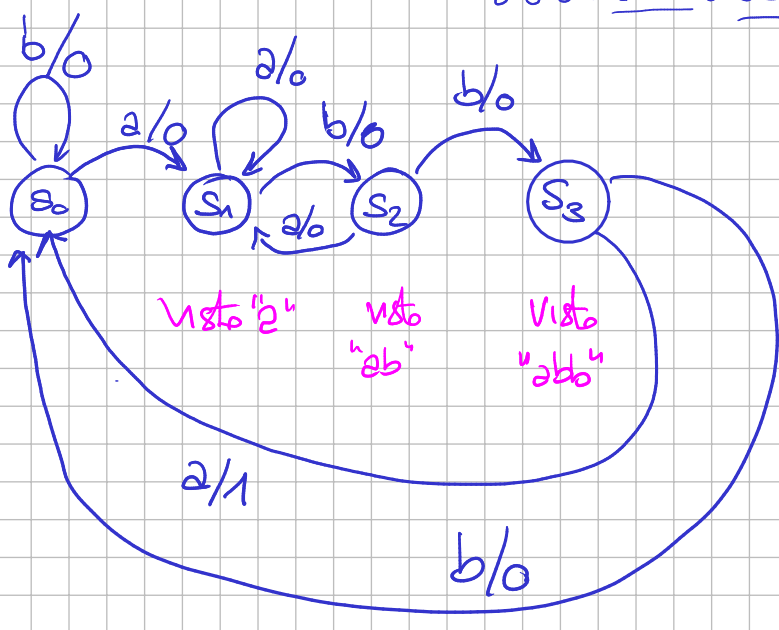
k	0	0000
	1	0001
	2	
	3	1111
	4	

$\{a, b\}$

abbaa

abbaaabbbaabbaa^{bb}
 0000000000000001

1)



- S₀ 00
- S₁ 01
- S₂ 10
- S₃ 11

a=0
b=1

$$z = r_1 r_0 \bar{x}$$

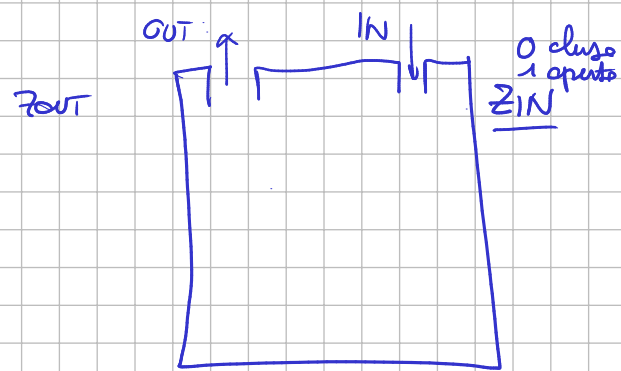
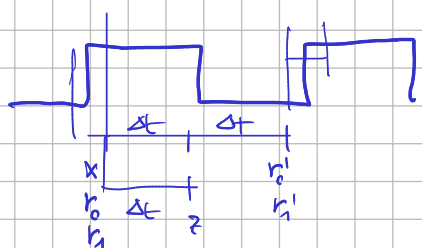
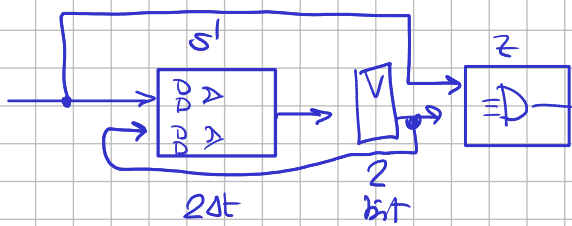
$$r_1' = \bar{r}_1 r_0 x + r_1 \bar{r}_0 x$$

$$r_0' = \bar{r}_1 \bar{x} + r_1 r_0$$

3)

state
2 bit
 $\lceil \log_2 4 \rceil = 2$
 $\swarrow \searrow$
 S₀ S₁ S₂ S₃

s	x		
r ₁ r ₀	x	z	r ₁ ' r ₀ '
00	0	0	01
	1	0	00
01	0	0	01
	1	0	10
10	0	0	01
	1	0	11
11	0	1	00
	1	0	00



150 posti

